Reservoir computing

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1 State of the art in reservoir computing

1.1 ESN - Echo State Network

1.1.1 Principles of reservoir computing

Based on the [1]: The state of reservoir dynamics can be expressed as:

$$h_t = f\big((1-k)\cdot u_t \cdot W_{in} + k \cdot h_{\mathrm{t-1}} \cdot W_h + y_{\mathrm{t-1}} \cdot W_{1b} + b\big) \tag{1}$$

Where:

h_{t-1} – are the reservoir state respectively, from the previous time step,
u_t – is the observed data at time step t,

y_{t-1} – is the the predicted output state $t-1$,
$W_{in} \in \mathbb{R}^{N_u \times N_h}$ – is the input weight matrix,
$W_h \in \mathbb{R}^{N_h \times N_h}$ – is the internal weight matrix,
$W_{1b} \in \mathbb{R}^{N_h \times N_y}$ – is the output fedback weight matrix,
$b \in \mathbb{R}^{N_h}$ – is the bias vector.
f – is the activation function, typically tanh or sigmoid,
k – is the leaking rate, typically $k \in [0.1, 0.3]$.

With the computed reservoir dynamics, the output can be then obtained by:

$$y_t = h_t \cdot W_{\text{out}} \tag{2}$$

Where:

$W_{\text{out}} \in \mathbb{R}^{N_h \times N_y}$ – is the output weight matrix.

1.1.2 Echo state property

Any system that changes in a nonlinear way can work as the reservoir. However, starting a nonlinear system with random connection strengths creates problems. The reservoir is a system that feeds its outputs back into itself. This can make it unstable if the connection strengths aren't set up correctly. For example, if the internal connections are too strong, the system might get stuck giving the same output regardless of what input it receives. The random connection strengths must be chosen so the system doesn't grow out of control. For the system to work well, it must follow something called the "echo state property." This rule ensures that the reservoir's behavior eventually depends on the input signal rather than just its starting conditions. To meet this requirement, the internal connection matrix W_h is first set up using random values between -1 and 1. This matrix is then adjusted one time according to the echo state property rule:

$$W_h' = \alpha \odot W_h \tag{3}$$

$$W_h^{\dagger} = \frac{\rho W_h}{|\lambda_{max}(W_h)|} \tag{4}$$

Where:

$\rho \in (0, 1)$ – is the spectral radius, typically $\rho \in [0.9, 1]$
$\lambda_{max}(W_h)$ – is the largest eigenvalue of W_h.
$\alpha \in (0, 1)$ – is the sparsity coefficient, typically $\alpha \in [0.1, 0.3]$.

The spectral radius is a parameter that determines the amount of nonlinear interaction of input components through time.

Due to the recursive nature of the reservoir layer, such dynamics reflect trajectories of the past historical input—the short-term memory (known as the fading memory). As another critical property for computing the RC principle, short-term memory can be quantitative measured by the coefficient of memory capacity

$$MC = \sum_{k=1}^{\infty} MC_k = \sum_{k=1}^{\infty} d^2(u_{t-k}, y_t) = \sum_{k=1}^{\infty} \frac{\text{cov}^2(u_{t-k}, y_t)}{\sigma^2(u_t)\sigma^2(y_t)} \tag{5}$$

Where:

$d^2(u_{t-k}, y_t)$ – is the square of the correlation coefficient between the output y_t and the input u_{t-k} with a delay of k time steps,

According to the Lyapunov stability analysis, a large memory capacity is needed to compute the RC principle, which can be achieved at the asymptotically stable region.

1.1.3 Learning algorithm

The training of the reservoir computing model involves adjusting only the output weights W_{out}. The input weights W_{in}, internal weights W_h, and feedback weights W_{1b} are typically initialized randomly and remain fixed during training. The training process can be summarized in the following steps:

$$Y = H \cdot W_{out} \tag{6}$$

Where:

$Y \in \mathbb{R}^{T \times N_y}$ – is the matrix of target outputs for all time steps,
$H \in \mathbb{R}^{T \times N_h}$ – is the matrix of reservoir states for all time steps,
T – is the total number of time steps.

In general, the W_{out} can be directly obtained by calculating the Moore-Penrose pseudoinverse of the reservoir states matrix H with respect to the target outputs matrix Y:

$$W_{out} = Y \cdot H^T \cdot (H \cdot H^T + \eta I)^{-1} \tag{7}$$

Where:

H^T – is the transpose of matrix H,
η – is the regularization parameter, typically $\eta \in [10^{-6}, 10^{-2}]$,
I – is the identity matrix of size $N_h \times N_h$.

1.1.4 Model development

The echo state network (ESN) and the liquid state machine (LSM) are the two representations of RC. While ESN and LSM are topographically equivalent, the former adopts the actual numerical data from the input to compute the network dynamics, and the latter adopts the spiking signal to represent the spatiotemporal pattern. **The LSM can be also seen as a SNN, where the reservoir has numerous leaky integrate-and-fire (LIF) neurons interconnected with the same recursive nature as in ESN**. As the spiking event is involved, the training of LSM generally relies on the spike-timing-dependent plasticity (STDP) and short-term plasticity (STP), among other training methods for spiking networks.

1.2 Time-delayed reservoir computing

Based on [5, 9, 8].

The model of reservoir can be written in the following general form:

$$\mathbf{r}(n) = F\big(\gamma \mathbf{W}_{in}\mathbf{x}(n) + \beta \mathbf{W}\mathbf{r}(n-1)\big), \tag{8}$$

$$\mathbf{y}(n) = \mathbf{W}_{out}\mathbf{r}(n). \tag{9}$$

Where:

$\mathbf{r}(n) \in \mathbb{R}^N$ – is the reservoir state at time step n,
$\mathbf{x}(n) \in \mathbb{R}^d$ – is the input vector at time step n,
$\mathbf{y}(n) \in \mathbb{R}^p$ – is the output vector at time step n,
$\mathbf{W}_{in} \in \mathbb{R}^{N \times d}$ – is the input weight matrix, usually drawn from a random distribution $[-1, 1]$,
$\mathbf{W} \in \mathbb{R}^{N \times N}$ – is the internal weight matrix, usually drawn from a random distribution $[-1, 1]$,
$\mathbf{W}_{out} \in \mathbb{R}^{p \times N}$ – is the output weight matrix,

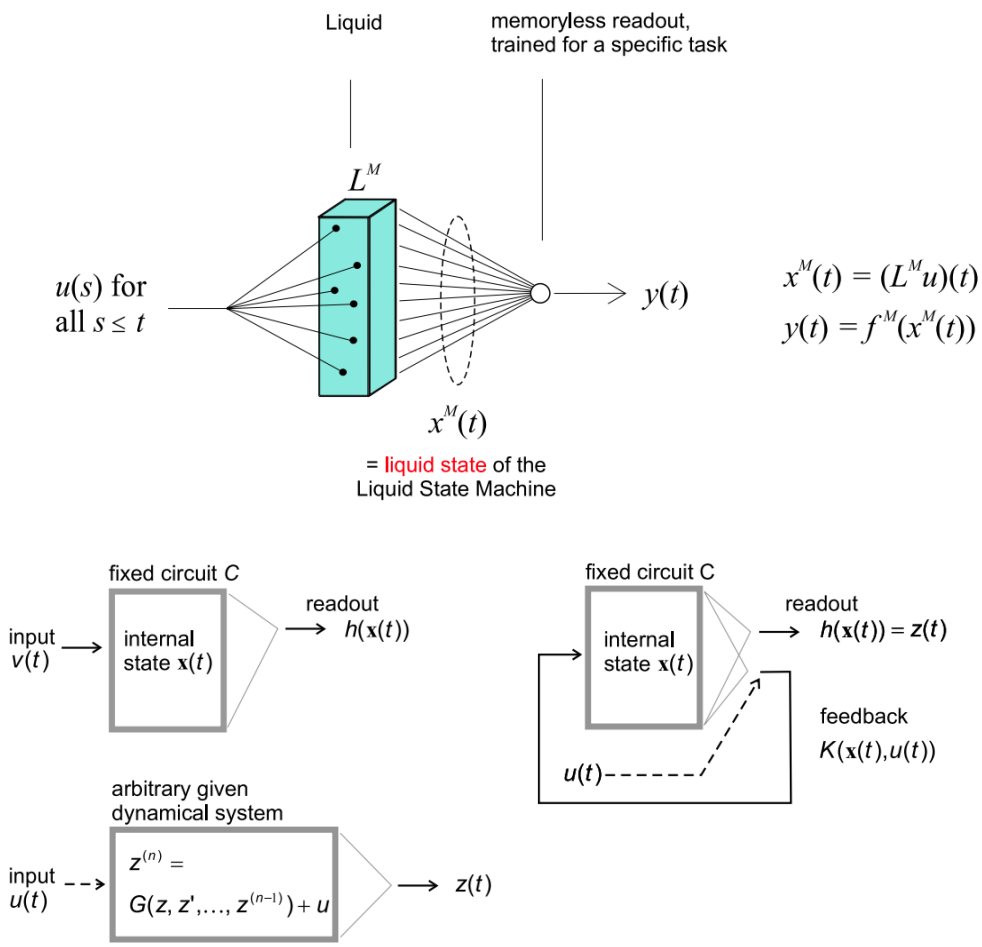


Figure 1: Liquid State Machine architecture

$\gamma, \beta \in \mathbb{R}^+$ – are the input and feedback scaling parameters,
$F : \mathbb{R}^N \to \mathbb{R}^N$ – is the nonlinear activation function, typically tanh or sigmoid.

In general, the time-delayed reservoir computing (TDRC) can be implemented using a single nonlinear node with delayed feedback loop instead of a large network of interconnected nodes. The delay dynamics create a high-dimensional state space that can be exploited for computation.

1.3 Liquid State Machine – Reservoir with spiking neurons

Based on [7, 2].

1.4 Next generation reservoir computing

Based on [4].

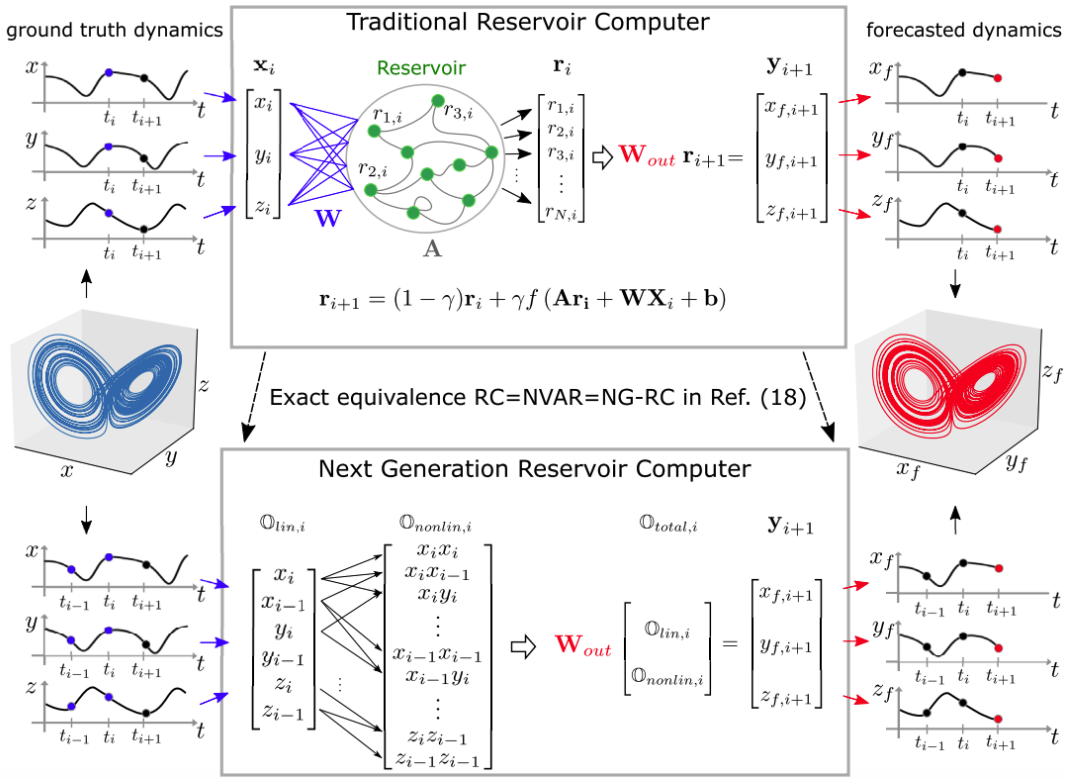


Figure 2: A traditional RC is implicit in an NG-RC. (top) A traditional RC processes time-series data associated with a strange attractor (blue, middle left) using an artificial recurrent neural network. The forecasted strange attractor (red, middle right) is a linear weight of the reservoir states. (bottom) The NGRC performs a forecast using a linear weight of time-delay states (two times shown here) of the time series data and nonlinear functionals of this data (quadratic functional shown here).

The reservoir is a dynamic system whose dynamics can be represented by:

$$\mathbf{r}_{i+1} = (1-\gamma)\mathbf{r}_i + \gamma f\big(\mathbf{A}r_i + \mathbf{W}_{in}\mathbf{X}_i + b\big) \tag{10}$$

Where:
$r_i = [r_i^1, r_i^2, \ldots, r_i^N]^T \in \mathbb{R}^{N\times 1}$ is the reservoir state at time step i,
The output layer expresses the RC output $\mathbf{Y}_{i+1}$ as a linear transformation of a feature vector $\mathbf{O}_{total,i+1}$, constructed from the reservoir state $\mathbf{r}_{i+1}$, through the relation:

$$\mathbf{Y}_{i+1} = \mathbf{W}_{out}\mathbf{O}_{total,i+1} \tag{11}$$

Where:

$\mathbf{O}_{total,i+1} \in \mathbb{R}^{M\times 1}$ – is the feature vector at time step $i+1$,
$\mathbf{W}_{out} \in \mathbb{R}^{p\times M}$ – is the output weight matrix,
M – is the size of the feature vector.

The RC is trained using supervised training via regularized least-squares regression. Here, the training data points generate a block of data contained in $\mathbf{O}_{total}$ and we match $\mathbf{Y}$ to the desired output $\mathbf{Y}_d$ in a least-square sense using Tikhonov regularization so that $\mathbf{W}_{out}$ is given by

$$\mathbf{W}_{out} = \mathbf{Y}_d\mathbf{O}_{total}^T(\mathbf{O}_{total}\mathbf{O}_{total}^T + \alpha\mathbf{I})^{-1} \tag{12}$$

A different approach to RC is to move the nonlinearity from the reservoir to the output layer. In this case, the reservoir nodes are chosen to have a linear activation function $f(\mathbf{r}) = \mathbf{r}$, while the feature vector $\mathbf{O}_{total}$ becomes nonlinear. A simple example of such RC is to extend the standard linear feature vector to include the squared values of all nodes, which are obtained through the Hadamard product $\mathbf{r} \odot \mathbf{r} = [r_1^2, r_2^2, \ldots, r_N^2]^T$

$$\mathbf{O}_{total} = \mathbf{r} \oplus (\mathbf{r} \odot \mathbf{r}) = [r_1, r_2, \ldots, r_N, r_1^2, r_2^2, \ldots, r_N^2]^T \tag{13}$$

Where:

$\oplus$ – is the concatenation operator.

Here, $\mathbf{O}_{total} = c \cdot \mathbf{O}_{linear} \oplus \mathbf{O}_{nonlinear}$, where c is a constant and $\mathbf{O}_{nonlinear}$ is a nonlinear part of the feature vector. Like a traditional RC, the output is obtained using these features.

The linear features $\mathbf{O}_{linear,i}$ at time step i is composed of observations of the input vector $\mathbf{X}$ at the current and at $k-1$ previous times steps spaced by s, where $(s-1)$ is the number of skipped steps between consecutive observations. If $\mathbf{X}_i = [x_{1,i}, x_{2,i}, \ldots, x_{d,i}]^T$ is a d-dimensional vector, $\mathbf{O}_{linear,i}$ has $d \cdot k$ components, and is given by

$$\mathbf{O}_{linear,i} = \mathbf{X}_i \oplus \mathbf{X}_{i-s} \oplus \mathbf{X}_{i-2s} \oplus \ldots \oplus \mathbf{X}_{i-(k-1)s} \tag{14}$$

Based on the general theory of universal approximators, k should be taken to be infinitely large. However, it is found in practice that the Volterra series converges rapidly, and hence truncating k to small values does not incur large error. This can also be motivated by considering numerical integration methods of ordinary differential equations where only a few subintervals (steps) in a multistep integrator are needed to obtain high accuracy. We do not subdivide the step size here, but this analogy motivates why small values of k might give good performance in the forecasting tasks considered below.

An important aspect of the NG-RC is that its warm-up period only contains sk time steps, which are needed to create the feature vector for the first point to be processed.

The nonlinear part $\mathbf{O}_{nonlinear}$ of the feature vector is a nonlinear function of $\mathbf{O}_{linear}$. While there is great flexibility in choosing the nonlinear functionals, we find that polynomials provide good prediction ability. Polynomial functionals are the basis of a Volterra representation for dynamical systems and hence they are a natural starting point. We find that low-order polynomials are enough to obtain high performance. All monomials of the quadratic polynomial, for example, are captured by the outer product $\mathbf{O}_{linear} \otimes \mathbf{O}_{linear}$, which is a symmetric matrix with $(dk)^2$ elements. A quadratic nonlinear feature vector $\mathbf{O}^{(2)}_{nonlinear}$, for example, is composed of the $(dk)(dk+1)/2$ unique monomials of $\mathbf{O}_{linear} \otimes \mathbf{O}_{linear}$, which are given by the upper triangular elements of the outer product tensor. We define $\lceil \otimes \rceil$ as the operator that collects the unique monomials in a vector. Using this notation, a p-order polynomial feature vector is given by

$$\mathbf{O}^{(p)}_{nonlinear} = \underbrace{\mathbf{O}_{linear} \lceil \otimes \rceil \mathbf{O}_{linear} \lceil \otimes \rceil \ldots \lceil \otimes \rceil \mathbf{O}_{linear}}_{p \text{ times}} \tag{15}$$

The total feature vector used for the Lorenz63 forecasting task is given by

$$\mathbf{O}_{total} = c \cdot \mathbf{O}_{linear} \oplus \mathbf{O}_{linear} \oplus \mathbf{O}^{(2)}_{nonlinear} \tag{16}$$

To allow the NG-RC to focus on the subtle details of this process, we use a simple Euler-like integration step as a lowest-order approximation to a forecasting step by modifying Eq. 2 so that the NG-RC learns the difference between the current and future step. To this end, Eq. 2 is replaced by

$$\mathbf{X}_{i+1} = \mathbf{X}_i + \mathbf{W}_{out} \mathbf{O}_{total,i} \tag{17}$$

2 Memristors in reservoir computing

2.1 Reservoir computing using dynamic memristors for temporal information processing

Based on [3, 6].

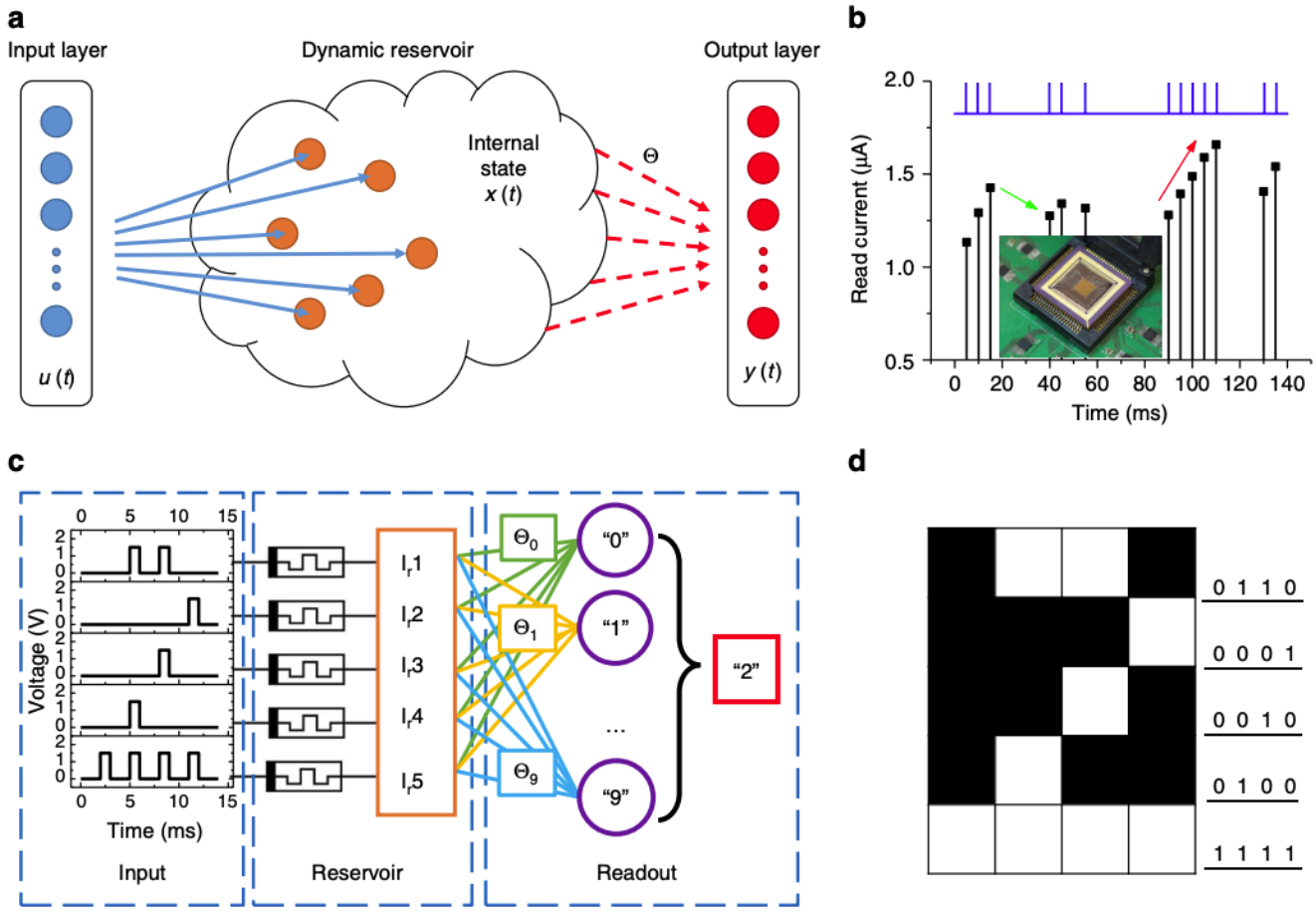


Figure 3: Reservoir computing system based on a memristor array. (a) Schematic of an RC system, showing the reservoir with internal dynamics and a readout function. Only the weight matrix Θ connecting the reservoir state x(t) and the output y(t) needs to be trained. (b) Response of a typical WOx memristor to a pulse stream with different time intervals between pulses. Inset: image of the memristor array wired-bonded to a chip carrier and mounted on a test board. (c) Schematic of the RC system with pulse streams as the inputs, the memristor reservoir and a readout network. For the simple digit recognition task of 5×4 images, the reservoir consists of 5 memristors. (d) An example of digit 2 used in the simple digit analysis.

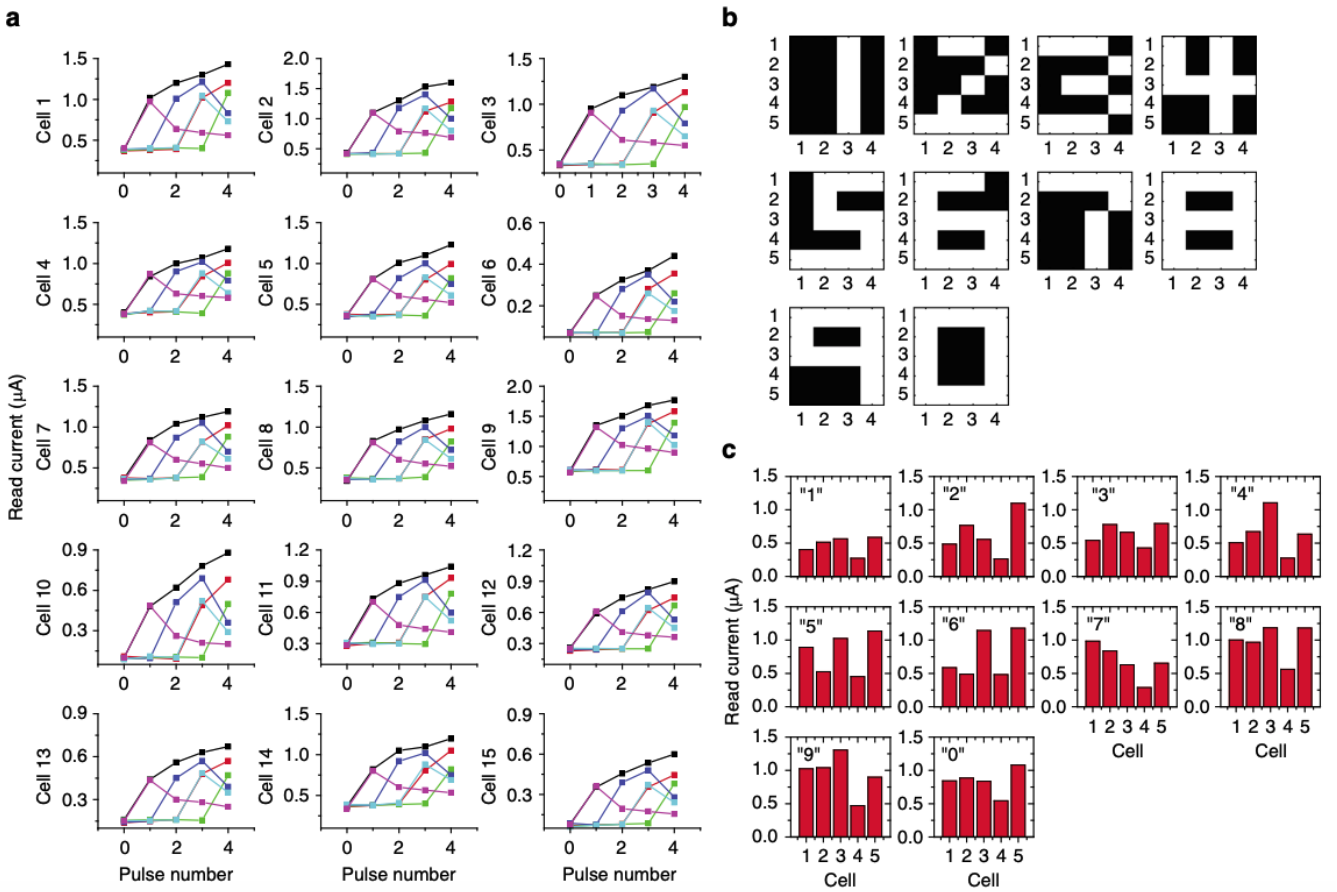


Figure 4: Reservoir states used to differentiate different temporal inputs. (a) The response of 15 memristors from the array to 6 different pulse streams (black: [1 1 1 1], purple: [1 0 0 0], blue: [0 1 1 0], red: [0 0 1 1], cyan: [0 0 1 0], green: [0 0 0 1]), showing similar response from all devices, as well as device variations. (b) Images of the 10 digits used in this test. (c) Experimentally measured reservoir states after the memristors are subjected to the 10 inputs. The reservoir state is reflected as the read currents of the 5 memristors forming the reservoir.

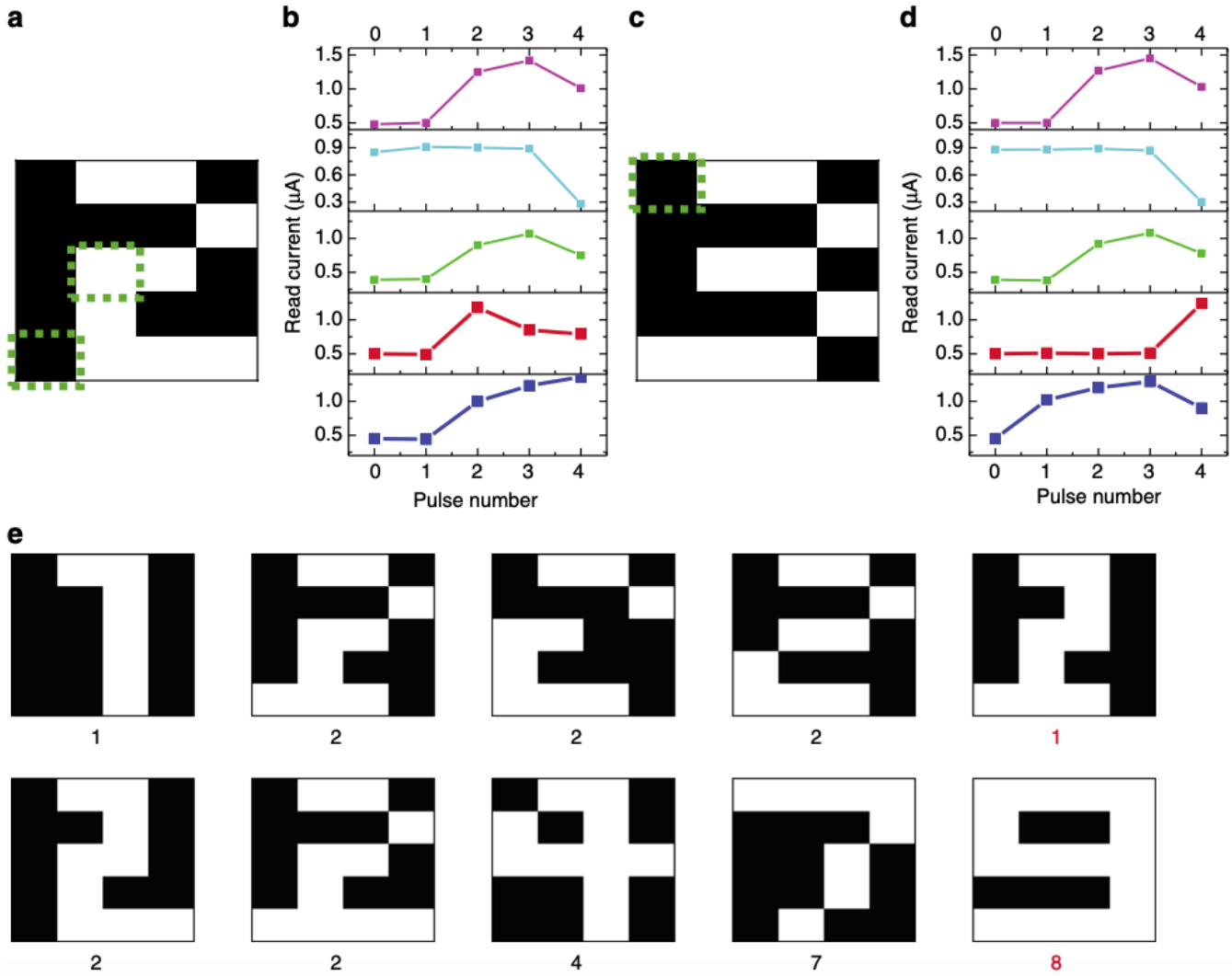


Figure 5: Recognition of noisy images. (a, c) Distorted images of digits 2 and 3 are generated by adding noise to the original data at locations marked by the dashed squares. (b, d) Corresponding reservoir states for these two inputs, showing differences in the two digits can be clearly captured by the memristors corresponding to the last two rows. (e) Recognition results of noisy digits. The RC system output, e.g., the predicted digit is shown below each case. The distorted images are generated by adding noise to the original training samples. The system can still successfully identify the majority of the distorted images without additional training, until too much noise is added as in the last two cases where the incorrect classifications are marked in red

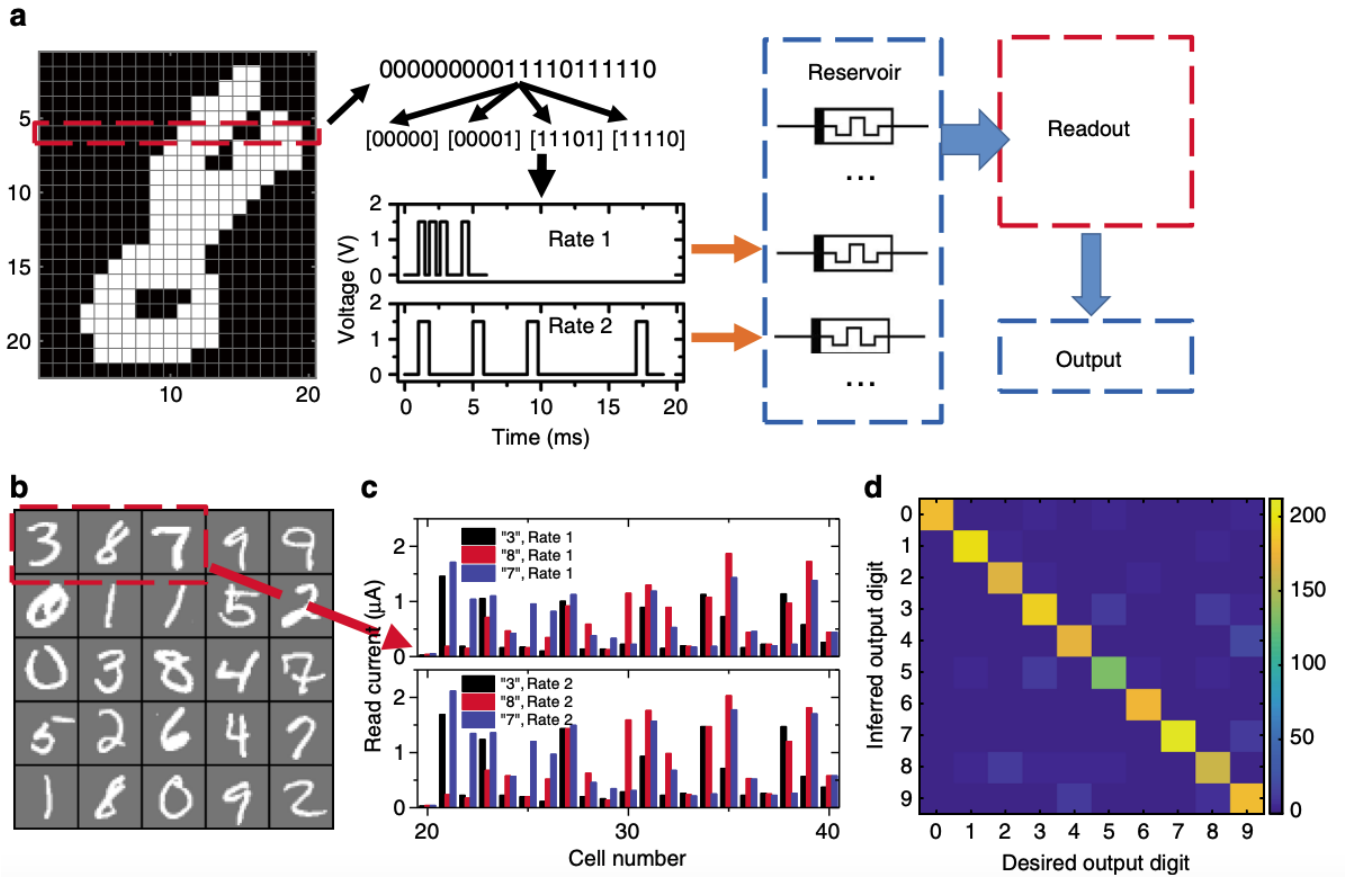


Figure 6: Handwritten digit recognition using a memristor-based RC system. (a) The process flow. The original digit image is first converted into pulse streams and fed to the memristor-based reservoir at different rates. The recognition result is generated after feeding the reservoir state to a trained readout function. (b) Some examples from the MNIST database. (c) Reservoir states corresponding to the three examples in (b) at two input rates (rate 1: timeframe width 0.8 ms, with pulse of 1.5 V, 0.5 ms; rate 2: timeframe width 4 ms, with pulses of 1.5 V, 0.8 ms), showing significant differences in the reservoir states. (d) False color confusion matrix showing the experimentally obtained classification results from the RC system vs. the desired outputs. The occurrence of the predicted output for each test case is represented by colors shown in the color scale. A recognition accuracy of 88.1% was obtained from the reservoir consisting of only 88 memristors.

3 Potential next steps in memristor-based reservoir computing

- Investigation of possible architectures for memristor-based reservoir computing systems, including hybrid approaches that combine memristors with other types of hardware.
- Development of training algorithms specifically designed for memristor-based reservoirs, taking into account the unique properties of memristors.
- Exploration of possibility of robust modelling of circuits with memristors within reservoir computing framework.

4 Experimental Investigation of Memristor-Based Reservoir Computing

The experimental investigation was conducted through computational simulations employing the Memristor Mean Metastable Switch (MMS) memristor model. The standardised MNIST (Modified National Institute of Standards and Technology) dataset, comprising 28×28 pixel grayscale images of handwritten digits, was used as the classification benchmark. This dataset provides a well-established baseline for evaluating neuromorphic computing architectures and enables direct comparison with prior work.

4.1 Row-Wise Input Encoding Methodology

The initial encoding strategy follows the row-wise decomposition proposed in [3]. Each 28×28 image is partitioned into 28 row vectors, which are presented sequentially to a reservoir consisting of 28 memristive elements, one per spatial position within the row.

Pixel intensities are encoded into the temporal domain via pulse amplitude modulation. Each grayscale value $I(i, j) \in [0, 255]$ is mapped to a voltage amplitude:

$$V_{\text{pixel}}(i, j) = V_{\max} \cdot \frac{I(i, j)}{255} \tag{18}$$

where $V_{\max}$ denotes the maximum voltage determined by the operational range of the memristive device.

4.1.1 Results

Under the row-wise encoding, the reservoir system achieved a classification accuracy of 60.37% on the MNIST test set. This relatively modest result reflects the inherent limitation of row-wise serialisation: each memristor processes a one-dimensional temporal signal devoid of local two-dimensional spatial context, constraining the representational capacity of the reservoir.

4.2 Patch-Based Feature Extraction for Memristor Crossbar Array Reservoir

To overcome the limitations of row-wise encoding and to exploit the parallel computational structure of memristor crossbar arrays, a patch-based spatial decomposition is applied to the input images prior to temporal encoding. This approach maps localised spatial features onto the row inputs of the crossbar, enabling simultaneous reservoir processing across all rows.

4.2.1 Spatial Patch Extraction

Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W}$, a sliding window of size $k_h \times k_w$ with stride s extracts a set of N patches $\mathcal{P} = \{P_1, P_2, \ldots, P_N\}$, where each patch $P_i \in \mathbb{R}^{k_h \times k_w}$ captures a localised spatial region. The patch at grid position (m, n) is given by:

$$P_{m,n} = \mathbf{I}\big[ms : ms + k_h - 1,\ ns : ns + k_w - 1\big] \tag{19}$$

with $m \in \{0, 1, \ldots, \lfloor (H - k_h)/s \rfloor\}$ and $n \in \{0, 1, \ldots, \lfloor (W - k_w)/s \rfloor\}$. For the MNIST images ($H = W = 28$) with kernel size $k_h = k_w = 8$ and stride $s = 3$, this yields a 7×7 grid of $N = 49$ patches per image.

4.2.2 Row-Wise Temporal Encoding for the Crossbar Array

Each extracted patch $P_i \in \mathbb{R}^{k_h \times k_w}$ is mapped onto the crossbar array such that each of the k_h rows of the patch drives a corresponding row input of the array. The r-th row of patch P_i, denoted

$\mathbf{p}_i^{(r)} = [p_{r,1}, p_{r,2}, \ldots, p_{r,k_w}]$, is encoded as a temporal sequence of k_w voltage pulses with amplitudes proportional to the pixel intensities:

$$V_r(t) = \sum_{c=1}^{k_w} p_{r,c} \cdot \Pi\left(\frac{t - (c-1)\,T_{\text{period}}}{D \cdot T_{\text{period}}}\right) \tag{20}$$

where $\Pi(\cdot)$ is the rectangular pulse function, T_{period} is the pulse period, and $D \in (0, 1]$ is the duty cycle. This encoding transforms each row of pixel values into an amplitude-modulated pulse train, preserving the spatial pixel ordering as a temporal sequence.

All k_h rows of a given patch are applied simultaneously, driving the corresponding row inputs of the crossbar in parallel. At each column c, the column wire accumulates current contributions from all k_h row memristors according to Kirchhoff's current law, so that the terminal state of the column memristor reflects a nonlinear integration over all row signals.

The patch extraction and crossbar encoding methodology is illustrated in Fig. 7. An example of the temporal pulse sequence applied to the first row of the crossbar is shown in Fig. 8.

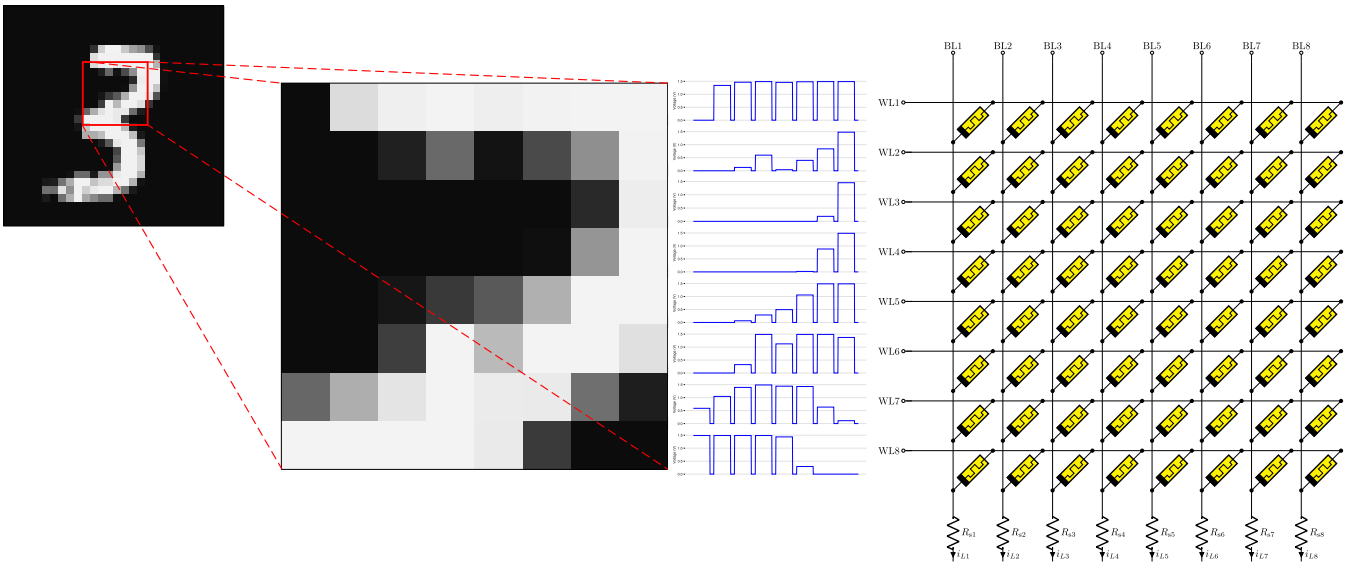


Figure 7: Patch-based feature extraction and encoding methodology for the memristor crossbar array reservoir. Each image is decomposed into overlapping patches, and each patch row is converted into a temporal pulse train applied to the corresponding crossbar row input.

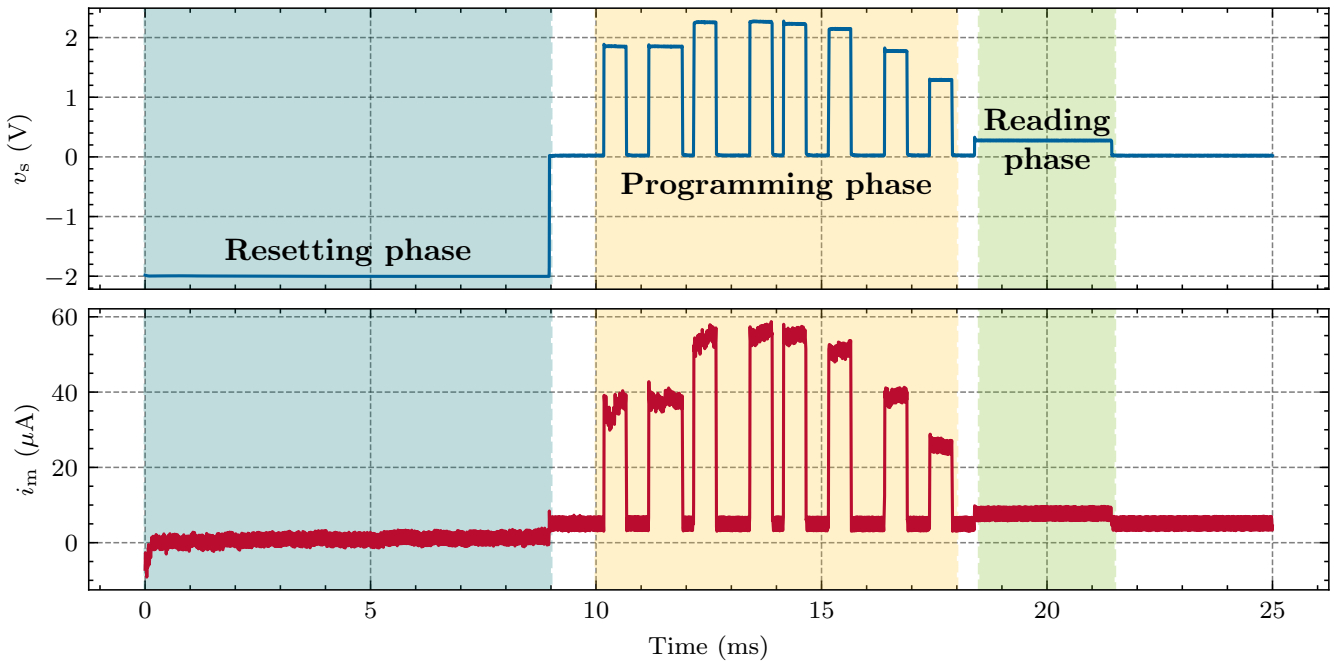


Figure 8: Example of the amplitude-modulated voltage pulse sequence applied to the first row of the memristor crossbar array during processing of a single patch.

4.2.3 Feature Vector Construction

After the complete pulse sequence for patch P_i has been applied, the terminal conductance states of all k_w column memristors are read out, yielding a local feature vector:

$$\mathbf{f}_i = \left[x_{i,1}^{\text{final}},\, x_{i,2}^{\text{final}},\, \ldots,\, x_{i,k_w}^{\text{final}}\right]^T \in \mathbb{R}^{k_w} \tag{21}$$

where $x_{i,c}^{\text{final}}$ denotes the terminal state of the memristor in column c after processing patch i. Each column state encodes a nonlinear transformation resulting from the simultaneous integration of all row signals at that column.

The global feature vector for a given image is formed by concatenating the local feature vectors from all N patches, with an appended bias term:

$$\mathbf{x} = \left[\mathbf{f}_1^T,\, \mathbf{f}_2^T,\, \ldots,\, \mathbf{f}_N^T,\, 1\right]^T \in \mathbb{R}^{Nk_w+1} \tag{22}$$

For the MNIST configuration ($N = 49$ patches, $k_w = 8$ columns), this yields a feature vector of dimensionality $49 \times 8 + 1 = 393$.

4.3 Classification Performance of the Patch-Based Reservoir

The patch-based reservoir was evaluated on the MNIST test set using ridge regression as the readout layer. The system achieved a classification accuracy of 87% in the experimental verification (compared to 91% obtained with the idealised simulation), representing a significant improvement over the 60.37% obtained with the row-wise encoding baseline. The 4 percentage point gap between simulation and experiment can be attributed to device-to-device variability, noise in the physical readout, and non-ideal switching dynamics of the fabricated memristors, which are not captured by the idealised MMS model.

This performance gain can be attributed to two complementary factors. First, the patch-based decomposition preserves local two-dimensional spatial correlations that are lost under row-wise

serialisation, allowing the reservoir to process spatially coherent inputs. Second, the crossbar architecture enables each column to aggregate nonlinear contributions from all rows simultaneously, producing a richer per-patch representation. The resulting 393-dimensional feature space provides the linear readout with substantially greater discriminative capacity compared to the 28-dimensional representation of the row-wise approach.

The confusion matrix (Fig. 9) shows high per-class accuracy across most digit categories, with the principal misclassifications occurring between structurally similar digit pairs such as $(3, 5)$ and $(4, 9)$. This pattern is consistent with the known morphological similarity structure of the MNIST dataset and indicates that the reservoir features capture meaningful shape distinctions.

The quality of the learned representations was further assessed by projecting the reservoir state vectors into two dimensions using t-SNE and UMAP (Fig. 10). Both embeddings exhibit well-separated, compact clusters corresponding to individual digit classes, confirming that the memristor reservoir maps the input patches into a feature space in which the class boundaries are approximately linearly separable.

These results demonstrate that memristor crossbar arrays, combined with a patch-based spatial encoding strategy, provide an effective physical substrate for reservoir computing in pattern recognition tasks. Future work will focus on optimising the patch extraction hyperparameters, exploring alternative temporal encoding schemes such as frequency modulation, and evaluating the scalability of the architecture to higher-resolution datasets and more complex classification problems.

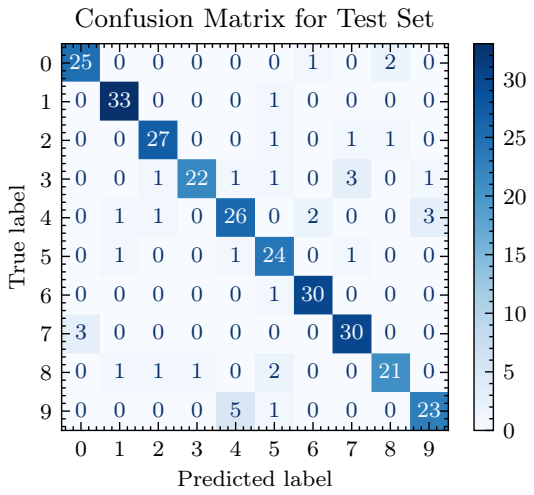
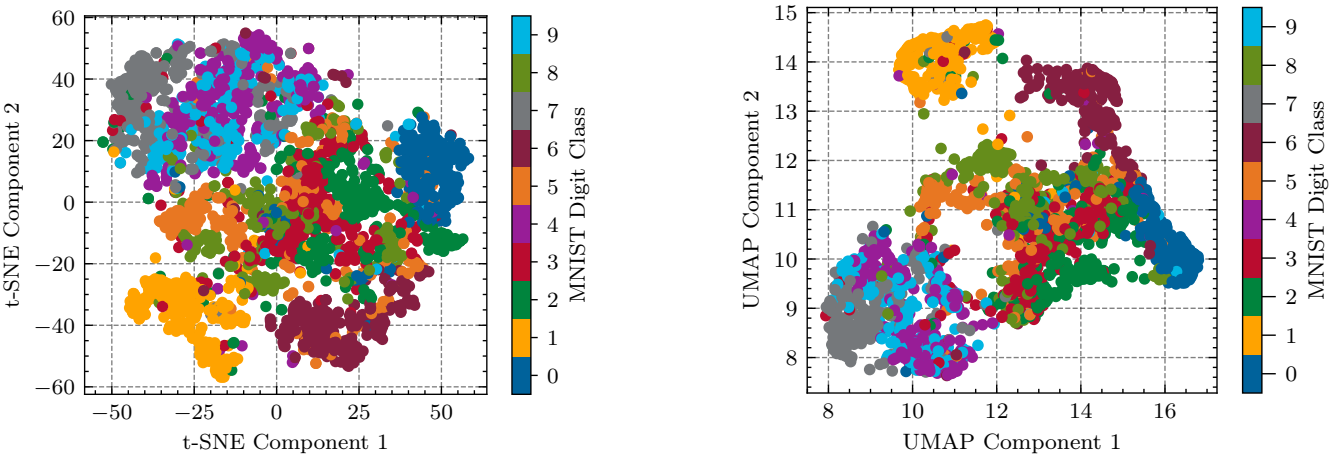


Figure 9: Confusion matrix for the patch-based memristor reservoir classifier on the MNIST test set. Diagonal elements indicate correct classifications; off-diagonal elements represent misclassifications.



(a) t-SNE projection

(b) UMAP projection

Figure 10: Two-dimensional embeddings of the reservoir feature vectors for the MNIST test set, computed via (a) t-SNE and (b) UMAP. Each point represents a single sample, coloured by digit class. The well-defined cluster structure confirms approximate linear separability of the reservoir representations.

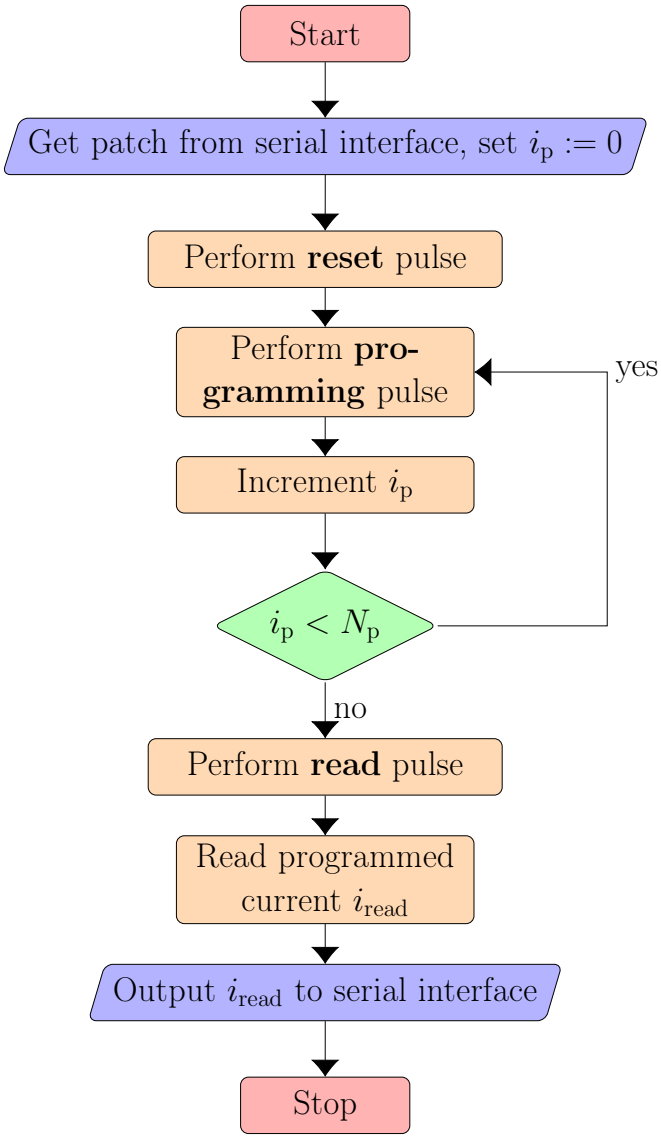


Figure 11: Flowchart of the memristor programming and readout procedure executed on the STM32 microcontroller for a single patch. Upon receiving a patch via the serial interface, the device iterates through N_p programming cycles, each consisting of a reset pulse followed by an amplitude-modulated programming pulse corresponding to the encoded pixel intensity. After all programming pulses have been applied, a read pulse is performed to measure the resulting memristor conductance state (i_{read}), which is transmitted back to the host via the serial interface.

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